

AMCSM – DEFENSIVE DISCLOSURE v2.0

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Repository: <https://github.com/blume-amcsm/amcsm>

Purpose of this disclosure

This Defensive Disclosure establishes the core concepts, system architecture and technical methods described herein as prior art, preventing third parties from filing patents or claiming exclusive rights on the disclosed ideas.

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Scope of the disclosure

This document discloses the foundational architecture of **AMCSM** (Advanced Musical Communication & Semantic Mapping) — a system designed for real-time, experience-based interaction between AI and musical environments.

Core Innovation

AMCSM introduces a **dual-model architecture** with a shared temporal foundation to enable true interactive learning in musical contexts:

1. **AMCSM-CORE** (LongBrain) A generative model responsible for producing musical output using MIDI 2.0 as its native communication protocol.
2. **AMCSM-SB** (ShortBrain) A smaller, online-adaptable auditor model that performs real-time analysis of the system's own output versus external feedback.
3. **Synchronized Cache Layer** Two time-synchronized ring buffers (Output-Cache and Input-Cache) that capture both the system's generated MIDI 2.0 events and incoming feedback with high-resolution timestamps.

Key Technical Concepts

- **MIDI 2.0 as semantic communication layer** The system treats MIDI 2.0 not merely as a control protocol, but as a rich, high-resolution semantic language for real-time musical interaction between AI and DAW / hardware controllers.
- **DAW as Master Clock** The Digital Audio Workstation provides the global time reference (MIDI Clock + high-resolution timestamps). All generated and received events are anchored to this common timeline.

- **Temporal Experience Learning (TEL)** The system records its own actions (Output) and the resulting reactions from the environment (Input) on the same time axis. The ShortBrain identifies deviations and creates labeled experience pairs, which are later used by the CORE model for causal learning and refinement.
- **Property Exchange & Musical Co-Regulation** Bidirectional exchange of musical parameters and expressive properties between human musician and AI through MIDI 2.0, enabling dynamic co-creation and mutual adaptation.
- **Separation of Concerns** Clear separation between real-time generation & observation (ShortBrain + Cache) and long-term consolidation & identity building (CORE).

Note

This disclosure describes the **conceptual architecture and core mechanisms**. Detailed implementations, model specifications, training procedures and source code will be published incrementally in the associated GitHub repository.

Legal Effect

By publishing this document, the described concepts, architecture and methods are established as **prior art** as of 23.05.2026. They:

- cannot be exclusively patented,
 - remain freely usable by the public,
 - are compatible with open-source development.
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Signature

I hereby confirm the publication of this Defensive Disclosure.

Signature: 

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